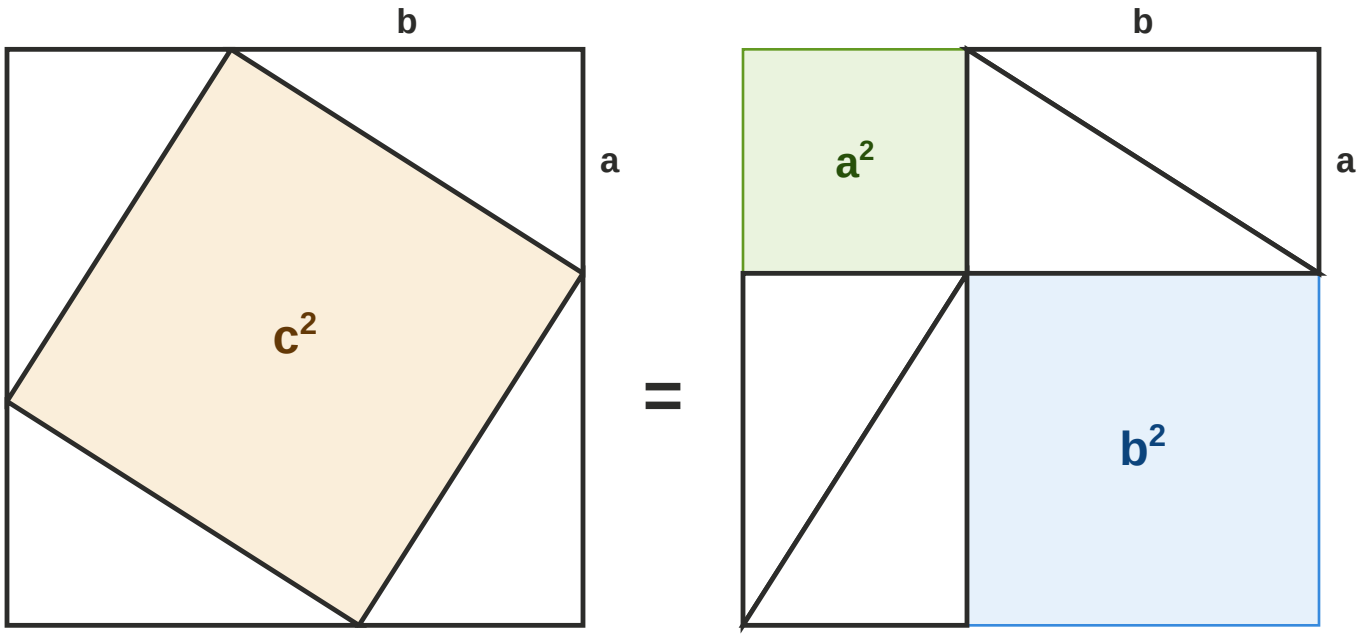
Why the Pythagorean theorem works

A geometric proof using four identical right triangles

Arrangement 1 **Arrangement 2**



Leftover: c^2 Leftover: $a^2 + b^2$

Same big square, same four triangles, so the leftover areas match:

$$a^2 + b^2 = c^2$$

How to read the picture

1. Take a right triangle with short sides *a* and *b* and hypotenuse *c*. Make four identical copies.
2. Place them inside a square of side $a + b$ so they leave a tilted square on the hypotenuse. Its side is *c*, so the leftover area is c^2.
3. Slide the same four triangles into new positions, without rotating them, so they leave two squares instead: one with side *a* and one with side *b*. The leftover area is now $a^2 + b^2$.
4. The big square has not changed and the four triangles have not changed, so what is left over must be the same size in both. Therefore $a^2 + b^2 = c^2$.

This works for any right triangle, because the argument uses only the letters a, b and c and no particular lengths.